

5.3.26 SPI characteristics

Unless otherwise specified, the parameters given in Table 66 are derived from tests performed under the ambient temperature, f_{PCLKx} frequency and supply voltage conditions summarized in Table 16.

- Output speed set to $OSPEEDR[1:0] = 11$
- Capacitive load $C_L = 30$ pF
- Measurement points done at $0.5 \times V_{DD}$ level
- I/O compensation cell activated

Refer to Table 46. I/O static characteristics for more details on the input/output alternate function characteristics (NSS, SCK, MOSI, MISO for SPI).

Table 66. SPI characteristics

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
f _{SCK}	SPI clock frequency	Master receiver mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	-	72	MHz
		Slave receiver mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	-	100	
		Slave transmitter mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	-	32	
t _{su} (NSS)	NSS setup time	Slave mode		3	-	-	ns
t _h (NSS)	NSS hold time	Slave mode		1	-	-	
t _w (SCKH)	SCK high and low time	Master mode		T _{SCK} ⁽¹⁾ - 1	T _{SCK} ⁽¹⁾	T _{SCK} ⁽¹⁾ + 1	
t _w (SCKL)							
t _{su} (MI)	Data input setup time	Master mode	DRDS = 0	2.5	-	-	
			DRDS = 1	13 - T _{SCK} ⁽¹⁾ /2	-	-	
t _{su} (SI)		Slave mode		2	-	-	
t _h (MI)	Data input hold time	Master mode	DRDS = 0	1	-	-	
			DRDS = 1	T _{SCK} ⁽¹⁾ /2 - 8	-	-	
t _h (SI)		Slave mode		1.5	-	-	
t _a (SO)	Data output access time	Slave mode		11	13.5	15.5	
t _{dis} (SO)	Data output disable time	Slave mode		8.5	10.5	12.5	
t _v (SO)	Data output valid time	Slave mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	12	15.5	
t _v (MO)	Data output valid time	Master mode		-	1.5	2	
t _h (SO)	Data output hold time	Slave mode (after enable edge)		8	-	-	
t _h (MO)	Data output hold time	Master mode (after enable edge)		0	-	-	

1. $T_{SCK} = T_{PCLK} \times \text{Baud rate prescaler}/2$.